

# Central Limit Theorem

## Motivation:

Sometimes we like to measure the characteristics of a population, but the population size is very large and that is not possible. In that case, we use sampling.

A sample statistics is a numerical value measured from sample and it is used to estimate the characteristics of the population (i.e. a parameter).

If we obtain a random sample and calculate a sample statistic from that sample, the sample statistic is also a random variable. The population parameters, however, are fixed. Since the sample statistics is a random variable, we should be able to find its distribution, mean and the standard deviation.

**Eg 1.** For example if we have the following distribution

| X        | 1   | 3   | 5   |
|----------|-----|-----|-----|
| $P_X(x)$ | 0.6 | 0.3 | 0.1 |

If we want to choose a sample of size = 5 from the above distribution, there are  $3^5 = 243$  different samples (with replacement) of size 5.

for example 1 1 3 3 5 is a sample of size 5 from the above distribution, another sample could be 3 1 1 3 5. We can write Generally like  $X_1, X_2, X_3, X_4, X_5$ , since for each it takes different values, so each one is like a random variable, and they all follow same distribution.

## Definition (Random Sample of IID)

A random sample of size  $n$  from a given distribution is a set of  $n$  independent r.v.'s  $X_1, X_2, \dots, X_n$ , each having the given distribution, with expectation  $E[X_i] = \mu$  and variance  $Var(X_i) = \sigma^2$ . Such a set of random variables is also called independent, identically distributed (iid or i.i.d).

## Definition (Sample sum)

Assume  $X_1, X_2, \dots, X_n$  is IID from a distribution with mean  $\mu$  and variance  $\sigma^2$  the sample sum is  $S = \sum_{i=1}^n X_i$

$$E[S] = n\mu \quad \text{and} \quad Var(S) = n\sigma^2$$

**Note:** If we need to Standardize the sample sum we should use

$$S^* = \frac{S - n\mu}{\sqrt{n}\sigma}$$

## Definition (Sample mean)

Assume  $X_1, X_2, \dots, X_n$  is IID from a distribution with mean  $\mu$  and variance  $\sigma^2$  the sample mean is denoted by  $\bar{X}$

$$\bar{X} = \frac{S}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

$$E[\bar{X}] = \mu \quad \text{and} \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

**Note:** If we need to Standardize the sample sum we should use

$$\bar{X}^* = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Intuitively: the Central Limit Theorem (CLT) says that the mean and the sum of a random sample of a large enough size from an (essentially) arbitrary distribution have approximately normal distribution:

## Central Limit Theorem

Assume  $X_1, X_2, \dots, X_n$  is IID from a distribution with mean  $\mu$  and variance  $\sigma^2$ , for large  $n$  we have:

(a) The sample sum  $S = \sum_{i=1}^n X_i$  is approximately normal  $N(n\mu, n\sigma^2)$ .

Equivalently, the standardized version  $S^* = \frac{S - n\mu}{\sqrt{n}\sigma} \approx N(0, 1)$

(b) The sample mean  $\bar{X}$  is approximately normal  $N(\mu, \frac{\sigma^2}{n})$ . Equivalently, the standardized version of  $\bar{X}^* = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \approx N(0, 1)$ .

# Combination of Normal random variables

- (a) Sums and differences of two independent normal r.v.'s: If  $X_1 \sim N(\mu_1, \sigma_1^2)$  and  $X_2 \sim N(\mu_2, \sigma_2^2)$  then,

$$X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

- (b) If each  $X_i \sim N(\mu_i, \sigma_i^2)$  and  $X_i$  are independent, then

$$\sum_{i=1}^n c_i X_i \sim N(c_i \mu_i, c_i^2 \sigma_i^2)$$

**Eg1-a.** A student commission wants to know the mean amount of money spent by college students for textbooks in one semester. Suppose the population mean is \$450 and the population standard deviation is \$40. A random sample of 625 students is taken.

What is the probability that the sample mean will be less than \$452?

$$\mu = 450, \quad \sigma = 40, \quad n = 625$$

$$\text{For large } n, \quad \bar{X} \propto N(\mu, \frac{\sigma^2}{n})$$

Recall for anything normally distributed we should first standardize.

$$P(\bar{X} < 452) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{452 - 450}{\frac{40}{\sqrt{625}}}\right) = P\left(Z < \frac{2}{\frac{40}{25}}\right)$$

$$P(\bar{X} < 452) = P\left(Z < \frac{50}{40}\right) = P(Z < 1.25) = \Phi(1.25)$$

Recall  $\Phi$  is cdf of standard normal.

**Eg1-b.** A student commission wants to know the mean amount of money spent by college students for textbooks in one semester. Suppose the population mean is \$450 and the population standard deviation is \$40. A random sample of 625 students is taken.

What is the probability that the sample mean will be within \$2 of \$450?

$$450 - 2 < \bar{X} < 450 + 2 \Rightarrow -2 < \bar{X} - 450 < 2$$

$$\begin{aligned} \Rightarrow |\bar{X} - 450| &< 2 \\ \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = Z & \\ \Rightarrow |Z| &< 1.25 \end{aligned} \quad \frac{|\bar{X} - 450|}{\frac{40}{25}} < \frac{2}{\frac{40}{25}}$$

$$\begin{aligned} P(|Z| < 1.25) &= \Phi(1.25) - \Phi(-1.25) \\ &= 1 - \Phi(-1.25) - \Phi(-1.25) = 1 - 2\Phi(-1.25) \end{aligned}$$


Eg2. The amount of sulfur in the daily emissions from a power plant has a normal distribution with mean of 134 pounds and a standard deviation of 22 pounds. For a random sample of 5 days, find the probability that the total amount of sulfur emissions will exceed 700 pounds.

Note: In this example  $n$  is small, we are allowed to assume  $S$  is normal since it is sum of normal random variables.

$$\mu = 134, \quad \sigma = 22, \quad n = 5$$

$$S = X_1 + X_2 + X_3 + X_4 + X_5 \quad S \sim N(n\mu, n\sigma^2)$$

$$P(S > 700) = P\left(\frac{S - n\mu}{\sqrt{n}\sigma} > \frac{700 - 5 \cdot 134}{\sqrt{5} \cdot 22}\right)$$

$$P(S > 700) = P(Z > 0.61) = 1 - P(Z \leq 0.61)$$

$$P(S > 700) = 1 - \Phi(0.61)$$

**Eg3.** An economist wishes to estimate the average family income in a certain population. The population standard deviation is known to be \$4500, and the economist uses a random sample of 225 families. What is the probability that the sample mean will fall within \$600 of the population mean?

$$\mu = ? \quad \sigma = 4500 \quad n = 225 \quad \tilde{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$P(|\bar{X} - \mu| < 600) = ?$$

$$P\left(\frac{|\bar{X} - \mu|}{\frac{\sigma}{\sqrt{n}}} < \frac{600}{\frac{4500}{\sqrt{225}}}\right) = P(|Z| < 2)$$



$$P(-2 < Z < 2) = 0.95$$



By 68- 95 - 99.7 rule